Summary-VIs

- **Use VIs to assist in image classification**
  - **To separate vegetated from non-vegetated areas**
  - **To distinguish between different types and densities of vegetation**
  - **To monitor seasonal variations in vegetative vigor**
  - **To map abundance, and distribution of vegetative vigor**
  - **To monitor health of vegetation**
  - **To monitor/estimate the growth and productivity of specific crops or of seasonal and yearly fluctuations in productivity**

Summary-VIs

- To compare ratios or indices over time or from one sensor to another,
  - Change digital values to radiances before calculating ratios, to account for differences in calibration of sensors

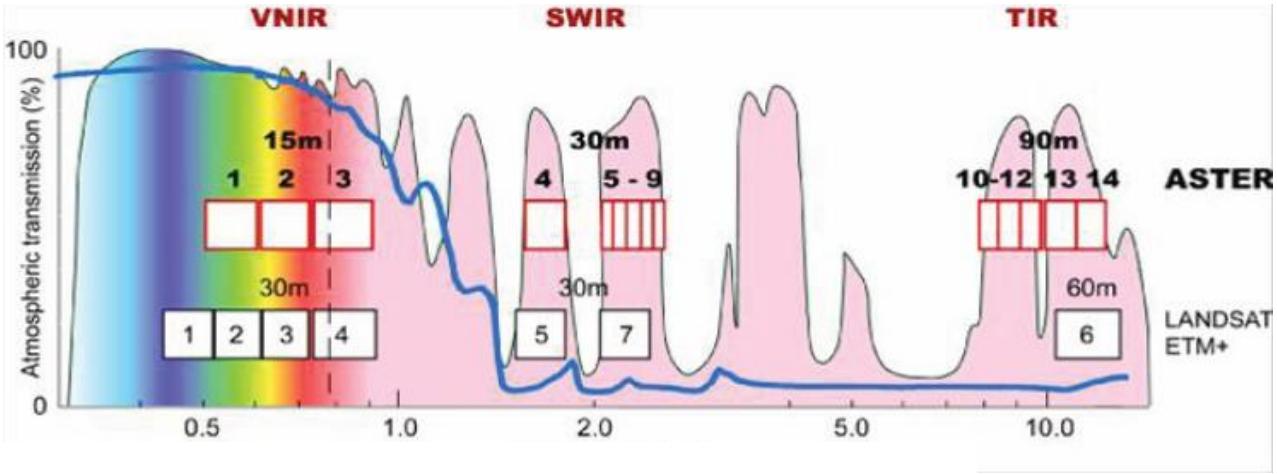
NDWI – Normalised Difference Water Index

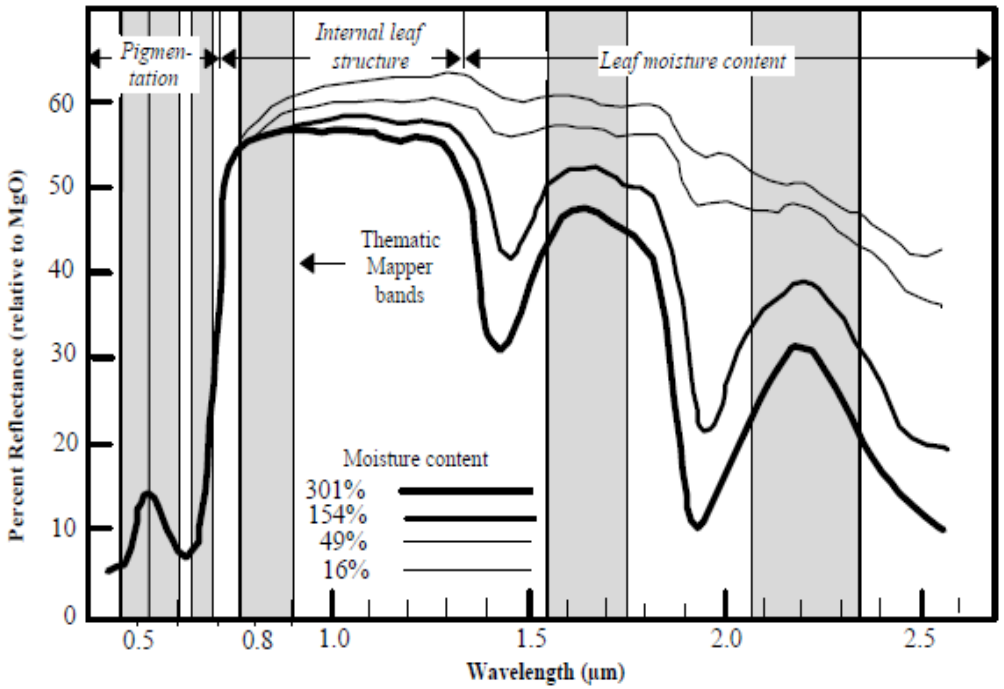
NDWI – Normalised Difference Water Index

- Dimensionless index with values ranging from -1 to 1
- It can be used to indicate the presence of water within the top layer of the soil (incase the soil is bare)
- It is calculated as the differences in the Near Infrared (NIR) and Short Wave Infrared (SWIR)

$$NDWI = \left(\frac{NIR - SWIR}{NIR + SWIR} \right)$$

NDWI



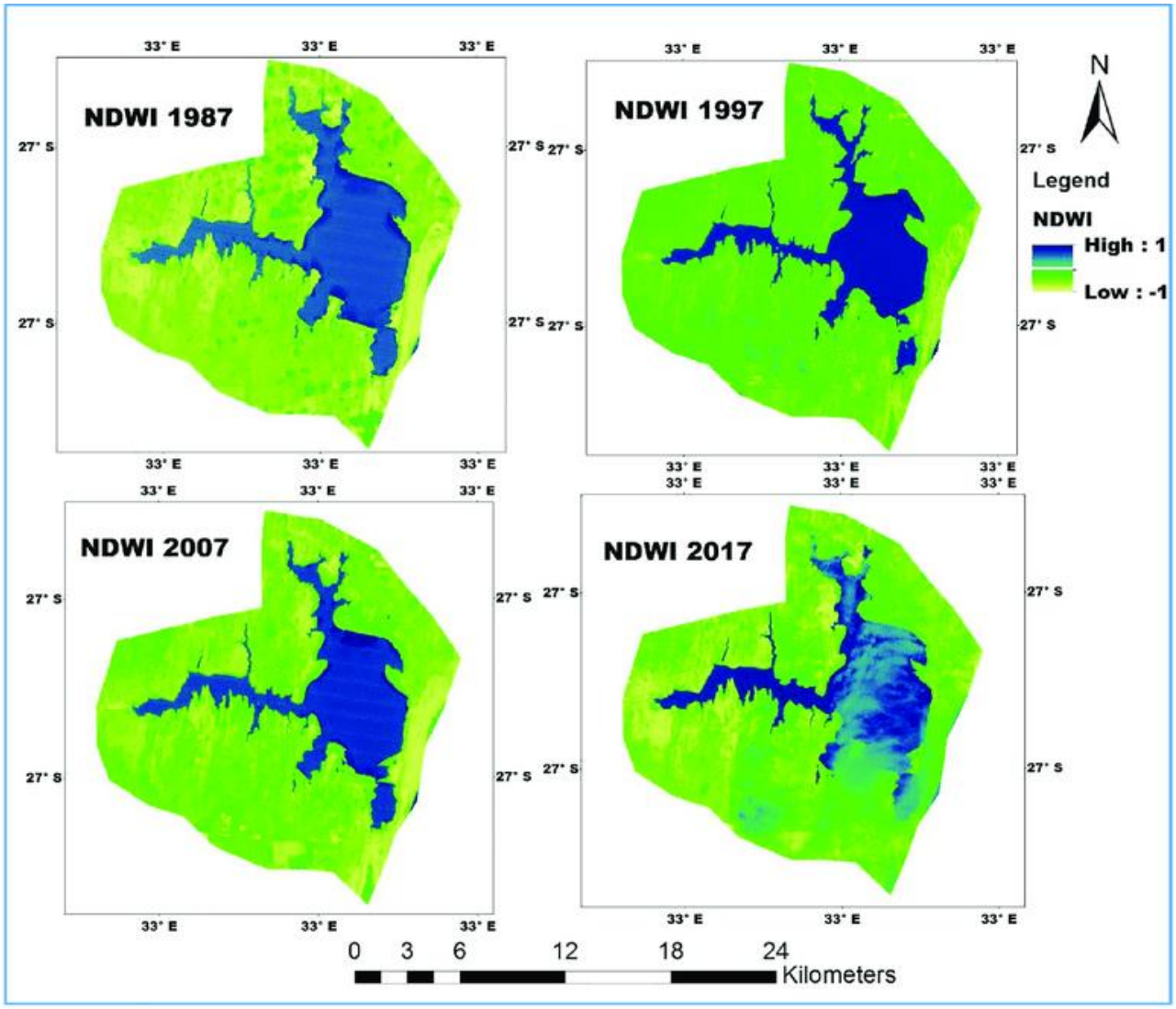


$$NDWI = \left(\frac{NIR - SWIR}{NIR + SWIR} \right)$$

$$NDWI = \left(\frac{NIR - GREEN}{NIR + GREEN} \right)$$

NDWI – Normalised Difference Water Index

- Time series analysis of NDWI of multi-temporal data



NDMI – Normalised Difference Moisture Index

Landsat 8

NDMI = (B5 - B6) / (B5 + B6)

(NIR - SWIR) / (NIR + SWIR)

MODIS

NDMI = (B2 - B6) / (B2 + B6)

$$Humidity = \left(\frac{NIR - MIR}{NIR + MIR} \right)$$

NDWI – Normalised Difference Snow Index (NDSI)

$$NDSI = \left(\frac{GREEN - IR}{GREEN + IR} \right)$$

Built-Up Indices

- **Normalized Difference Built-up Index**
  - NDBI = (SWIR – NIR) / (SWIR + NIR)
  - NDBI LS-8 = (B6 – B5) / (B6 + B5)
- **Normalized Built-up Area Index (NBAI)**
  - NBAI = (TM7 – TM5/TM2)/ (TM7 + TM5/TM2)

LS-8= (B7-B6/B2) /(B7+B6/B2)

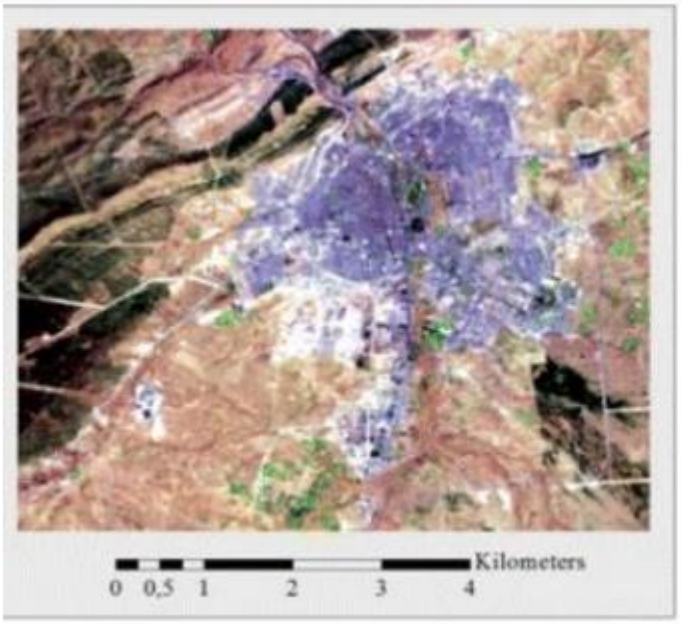
- **Band Ratio for Built-up Area (BRBA)**
  - BRBA = TM3 / TM5

LS-8= B4 /B6

- **New Built-Up Index (NBI)**
  - NBI=(TM3×TM5)/(TM4)

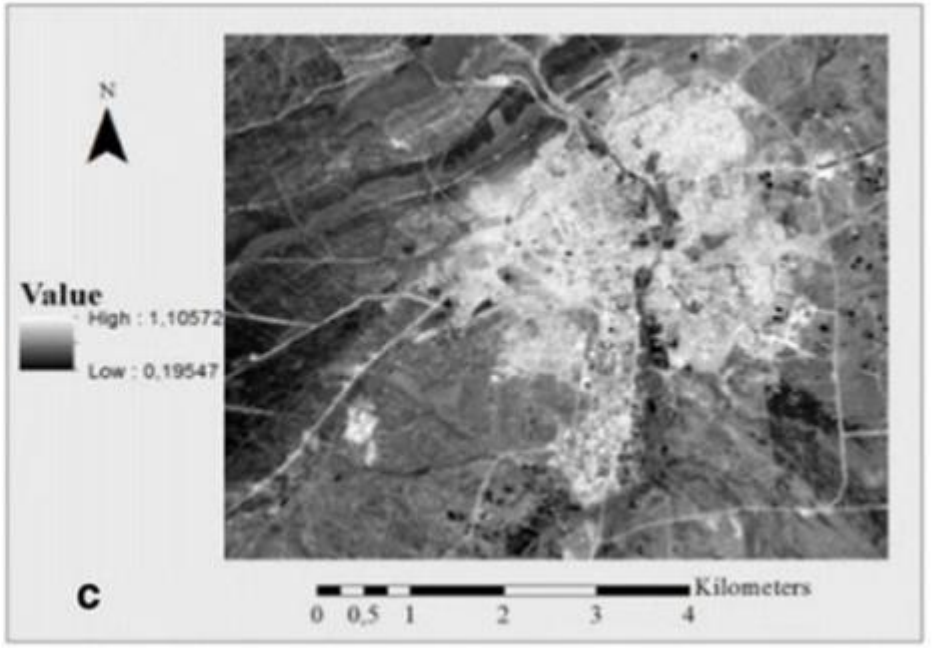
LS-8= B4x B6 /B5

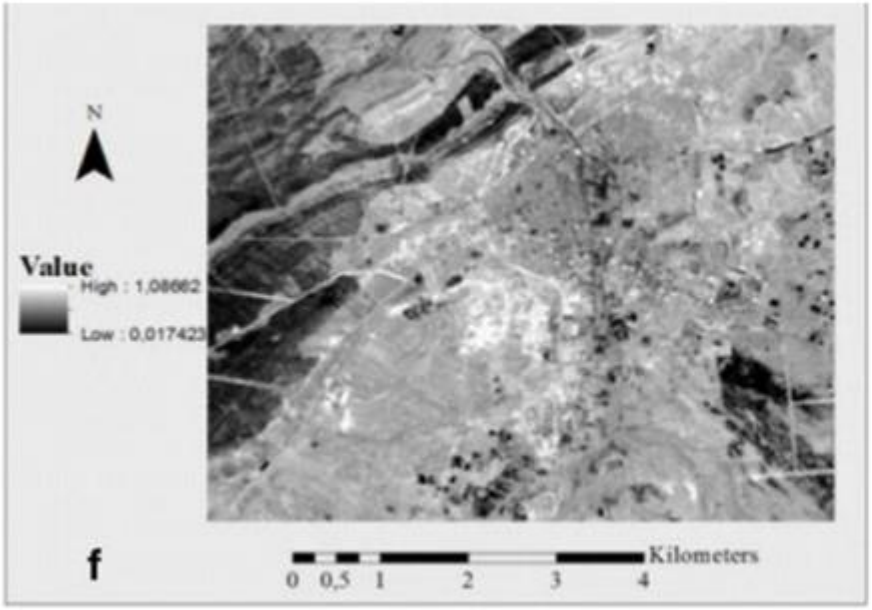
- One index developed to differentiate a particular urban area/city and bare soil may not necessarily apply to another area because spectral variance in response to soil and built-up surfaces



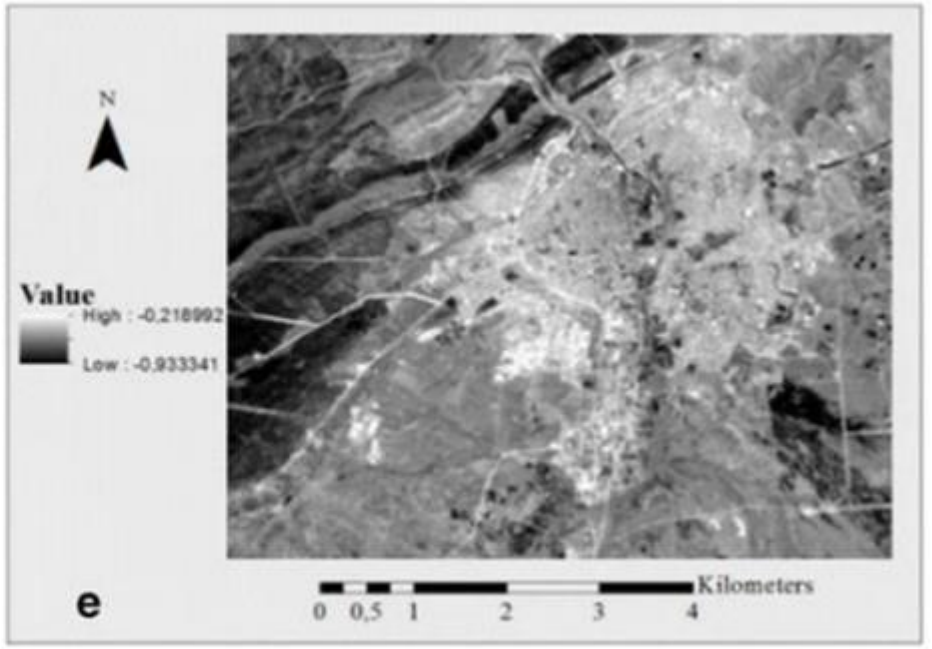
Image

BRBA





NBI



NBAI

Bare Soil Index (BSI)-Sentinel-2

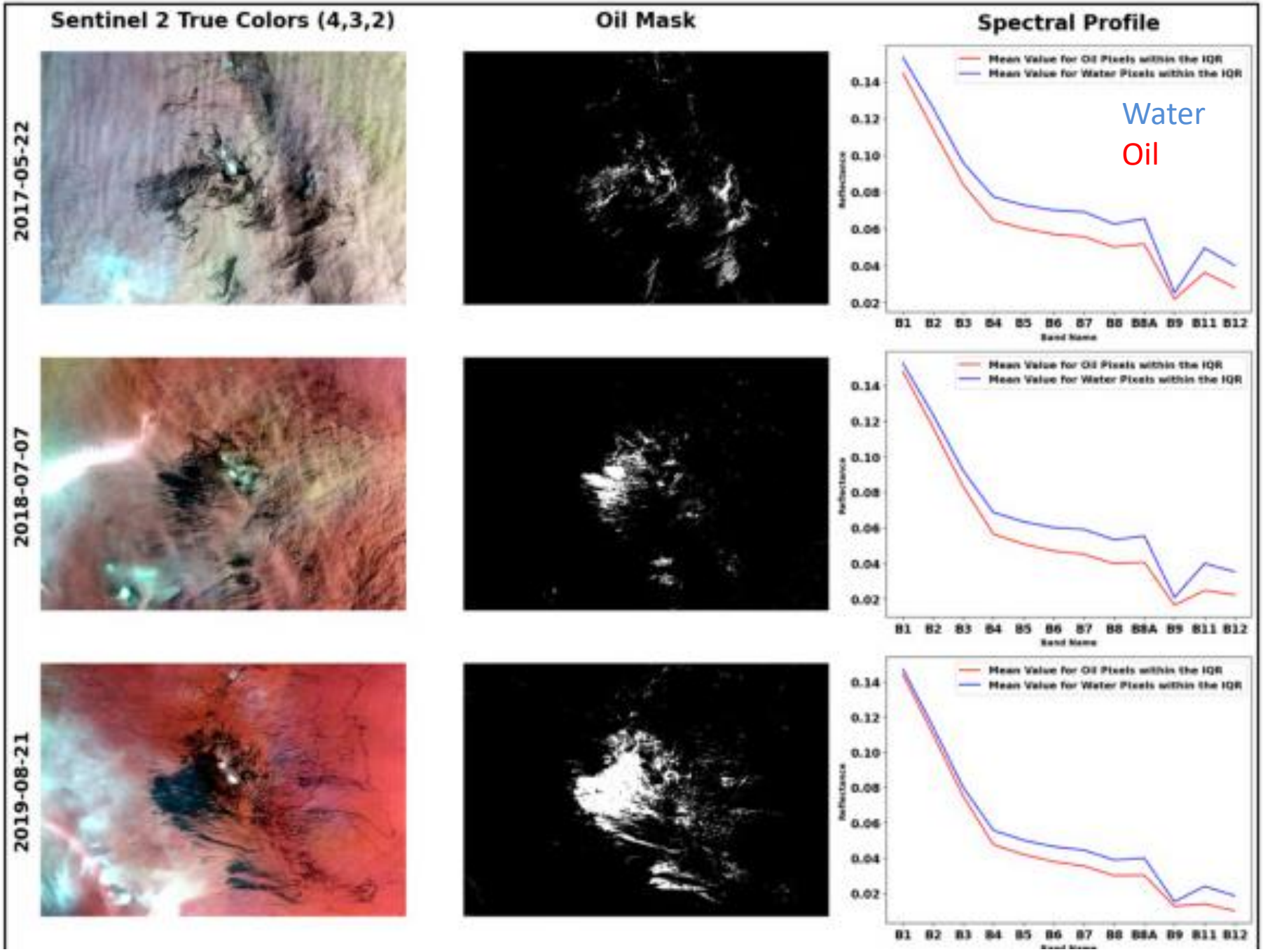
$$BSI = ((SWIR_1 + RED) - (NIR + BLUE)) / SWIR_1+RED) + (NIR +BLUE))$$

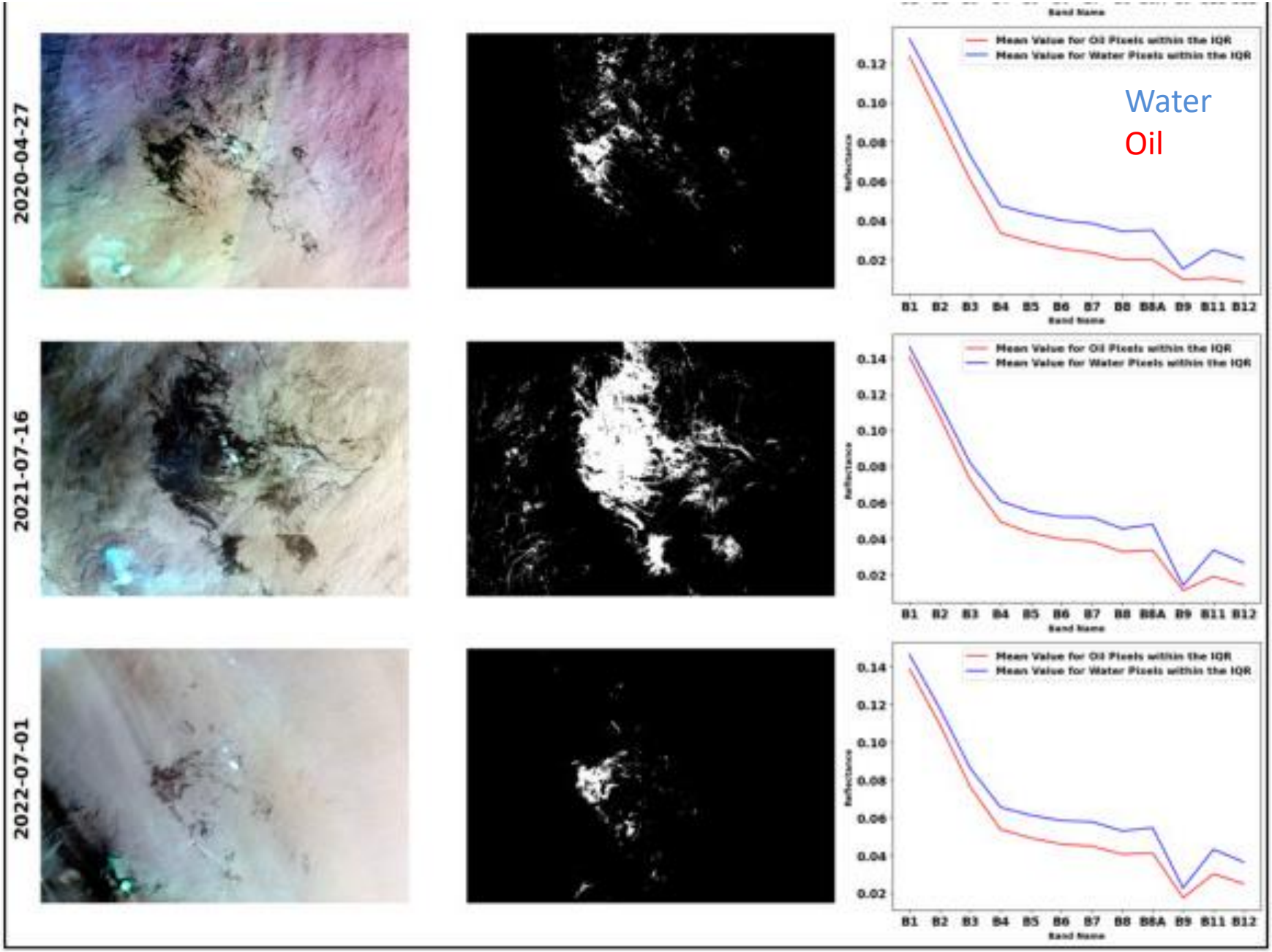
Modifies Bare Soil Index (MBI)-Landsat-8

$$MBI = (SWIR_1 - SWIR_2 - NIR) / (SWIR_1 + SWIR_2 + NIR) + 0.5$$

Sentinel-2 Based Indices

Index	**Mathematical expression**	**Sentinel-2 Bands formula**
NDVI	(NIR - Red) / (NIR + Red)	(B08 - B04) / (B08 + B04)
SAVI	((NIR – R)/(NIR + *R* + 0.5)) × (1 + 0.5)	((B08 – B04)/(B08 + B04 + 0.5)) × (1 + 0.5)
NDWI	(Green - NIR) / (Green + NIR)	(B03 - B08) / (B03 + B08)
NWI- New Water Index (NWI)	(Blue – SWIR1)/ (Blue + SWIR2)	(B02 – B12) / (B02 + B12)
MNDWI. Modified Normalized Difference Water Index	(Green – SWIR1) / (Green + SWIR1)	(B03 – B11) / (B03 + B11)
SWM Sentinel Water Mask	(Blue + Green)/(NIR + SWIR1)	(B02 + B3) / (B08 + B11)
NBR -Normalized Burn Ratio	(NIR – SWIR1) / (NIR + SWIR1)	(B08 – B11) / (B08 + B11)
OSI - Oil spill index	(Red + Green) / Blue	(B04 + B03) / B02


Sentinel 2 True Colors (4,3,2)
Oil Mask
Spectral Profile
2017-05-22
2018-07-07
2019-08-21
Mean Value for Oil Pixels within the IQR
Mean Value for Water Pixels within the IQR
Water
Oil
Reflectance
0.14
0.12
0.10
0.08
0.06
0.04
0.02
B1 B2 B3 B4 B5 B6 B7 B8 B8A B9 B11 B12
Band Name


2020-04-27
2021-07-16
2022-07-01
Mean Value for Oil Pixels within the IQR
Mean Value for Water Pixels within the IQR
Water
Oil
Reflectance
Band Name
B1 B2 B3 B4 B5 B6 B7 B8 B8A B9 B11 B12

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